



crambrunch
study smarter, manage better...

Crambrunch

study smarter , manage better

MEMBERS

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2. Ipsita Priyadarshini Prusty — Backend & Module Design- (23IMCA019)
3. Pujarani Sahoo — Database Management, Testing & Module Design – (23IMCA036)
4. Piyusa Pravas Priyadarshi — Documentation & Designing – (23IMCA030)



Introduction

“Crambrunch”- is a smart and web-based platform that is developed to organize and manage the academic as well as campus activities in a single secure system digitally. It aims to reduce the manual and paperwork by providing role-based access for students, teachers, parents and administrators. This ensures proper data management, transparency and improves administrative efficiency across educational institutions.

Why Crambrunch Matters :



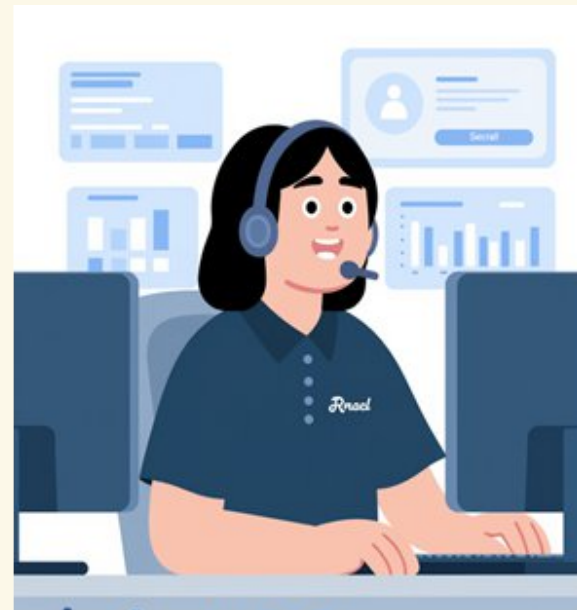
Student Management



AttenTrack



Feedback Control



Complaint Management



Skill Exchange

Objective

Reduce Manual Work

Establish Role-Based Access

Automate Academic Tracking

Enhance Communication



Increase Operational Efficiency

Increase Administrative Efficiency

Improve Campus Governance

Provide Feedback Mechanisms

System Comparison

Traditional ERP System

- **Static Attendance Logs:** Only record past absences without any forward-looking guidance for the student.
- **Identifiable Grievances:** Complaints are linked to user profiles, leading to fear of academic bias or retaliation.
- **Manual Substitution Workflow:** Relies on physical coordination, phone calls, or HOD intervention to fill empty slots.
- **Lack of Internal Accountability:** No transparent ledger to track who modified grades or sensitive records.
- **Passive Data Storage:** Acts as a digital filing cabinet rather than a proactive management tool.
- **Fragmented Career Support:** Focuses strictly on grades, ignoring practical skills or peer-to-peer learning.

CRAMBRUNCH

- **Instant Attendance Analyzer:** Smart attendance tracker with instant eligibility detection.
- **Anonymous Escalation Logic:** Protects student identity while ensuring tickets move to higher authorities if unresolved.
- **Automated Notification & Alerts:** Send alerts to students and parents when attendance drops below required percentage
- **Transparent Management System:** Provides role-based monitoring and controlled access for Admin, HOD, Teacher, Parent, and Student.
- **Real-time Substitution "Claiming":** Automates faculty gaps by allowing free teachers to instantly take over available periods.
- **Integrated Peer Exchange:** Facilitates a barter system for students to trade technical skills and knowledge.

Software Requirements :- (For Development)

COMPONENTS	TECHNOLOGY/TOOLS USED
Front-end	HTML, CSS, JS, React
Back-end	Nodejs, Express, JWT
Database	MongoDB
Hosting Site	Local Host
Code Editor	VS code

Hardware Requirements :- (For Development)

<u>COMPONENTS</u>	<u>REQUIREMENT</u>
Supported Devices	Desktop / Laptop / Tablet / Smartphone
Processor	Any modern processor(intel i3/i5 , AMD ryzen 3/5 or higher)
Operating System	Window / Mac OS / Linux / Android / iOS
RAM	Minimum 4 GB (8 GB recommended)
Storage	At least 16 GB free space
Internet Speed	Minimum 10 Mbps
Browser Support	Chrome, Edge (latest versions)

Module Description – Super Admin

The **Super Admin Module** provides complete authority over the entire platform with advanced administrative control and monitoring capabilities.

Features:

- Manage all admins and users.
- Monitor system-wide activities
- Handle escalated grievances and reports.
- Maintain database integrity and system performance
- Control overall platform configuration and permissions

Module Description – Admin

The **Admin Module** manages overall system operations and user activities while ensuring smooth platform functionality and security.

Features:

- Approve or reject user registrations.
- Manage students, teachers, and parent accounts.
- Monitor grievances and notifications.
- Control system data and academic records.
- Maintain platform security and accessibility.

Module Description – Teacher

The **Teacher** enables teachers to manage academic activities efficiently through attendance management, marks uploading, and material sharing.

Features:

- Mark and manage student attendance.
- Upload internal and semester marks.
- Share study materials and resources.
- Monitor student academic progress.
- View timetable and student feedback.

Module Description – Student

The **Student Module** provides students a personalized academic dashboard to monitor attendance, marks, grievances, study materials, and Skill Hub activities in real time.

Features:

- View real-time attendance percentage and eligibility status.
- Access marks and academic performance.
- Download study materials uploaded by teachers.
- Submit complaints/grievances anonymously.
- Participate in Skill Hub for skill exchange and task collaboration.

Module Description – Parent

The **Parent Module** allows parents to monitor their child's academic activities and attendance updates in real time.

Features:

- Track attendance and eligibility status.
- View marks and academic performance reports.
- Receive attendance warning alerts.
- Monitor student progress and notifications.

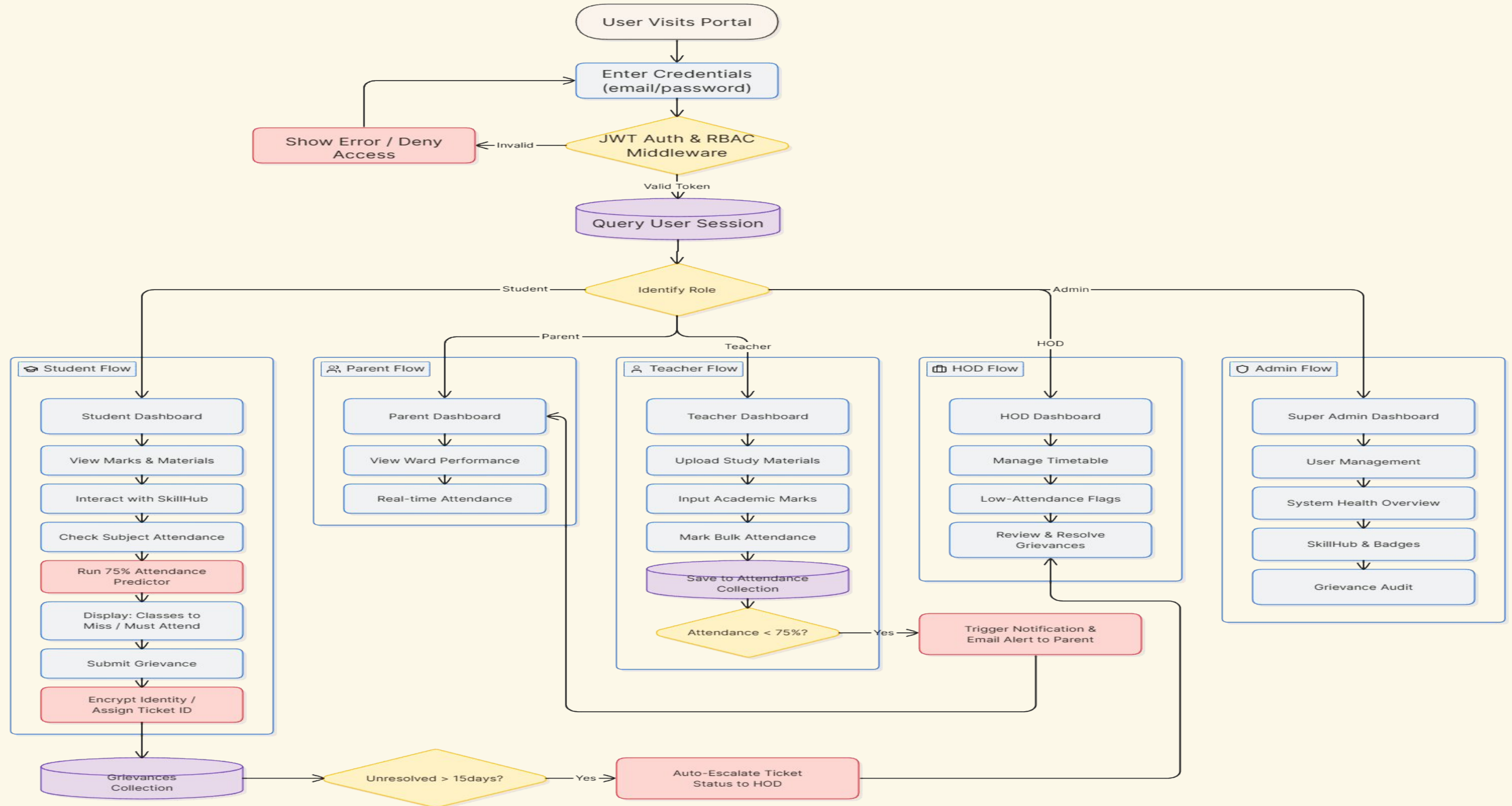
Module Description – SkillHub

The **SkillHub Module** promotes collaborative learning by allowing students to exchange skills, participate in tasks, and earn achievement badges.

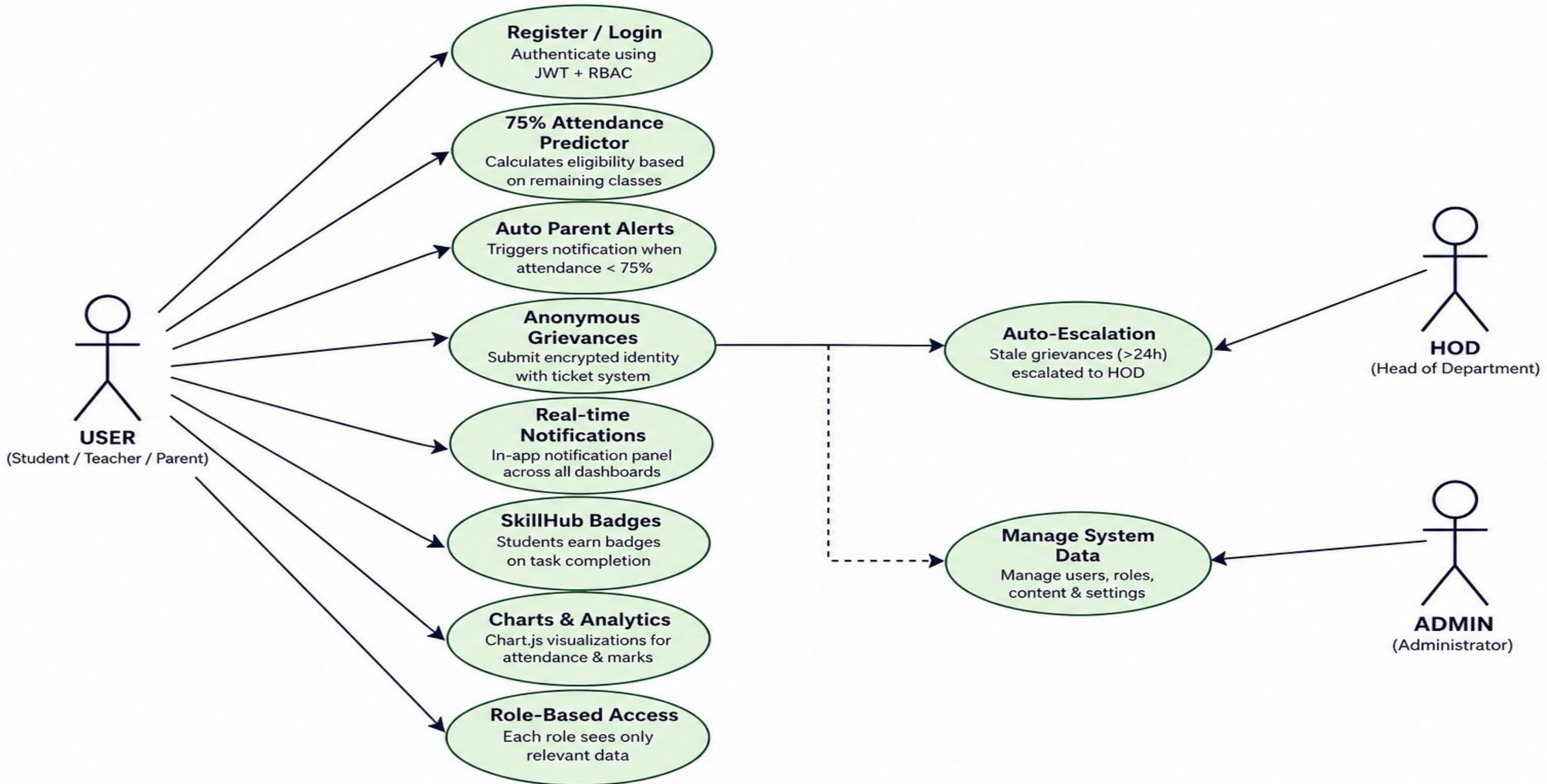
Features:

- Post offered and required skills.
- Participate in collaborative tasks.
- Earn badges and recognition.
- Improve peer-to-peer learning opportunities.

Flow Diagram

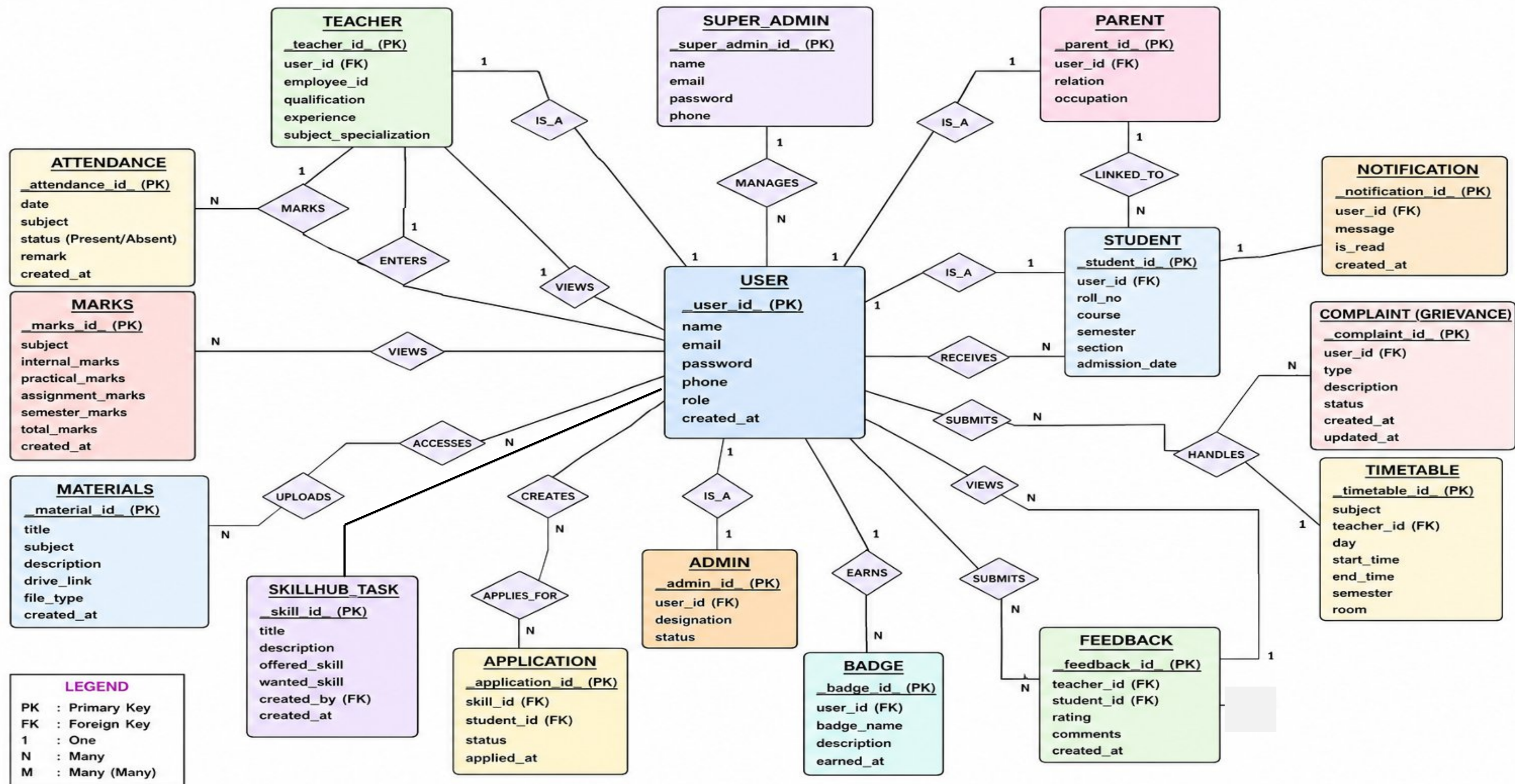


CRAMBRUNCH – Use Case Diagram

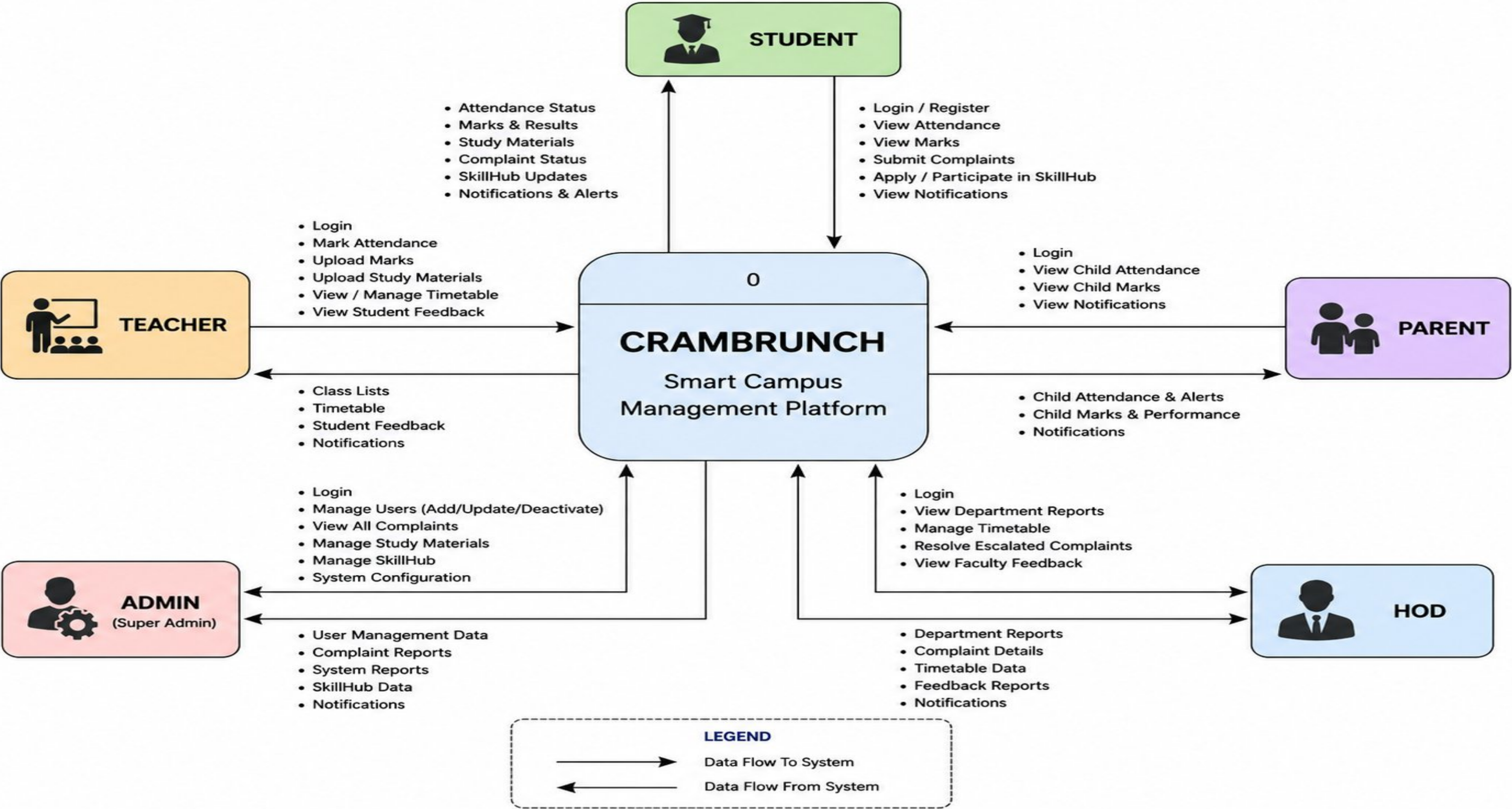


CRAMBRUNCH – SMART CAMPUS MANAGEMENT PLATFORM

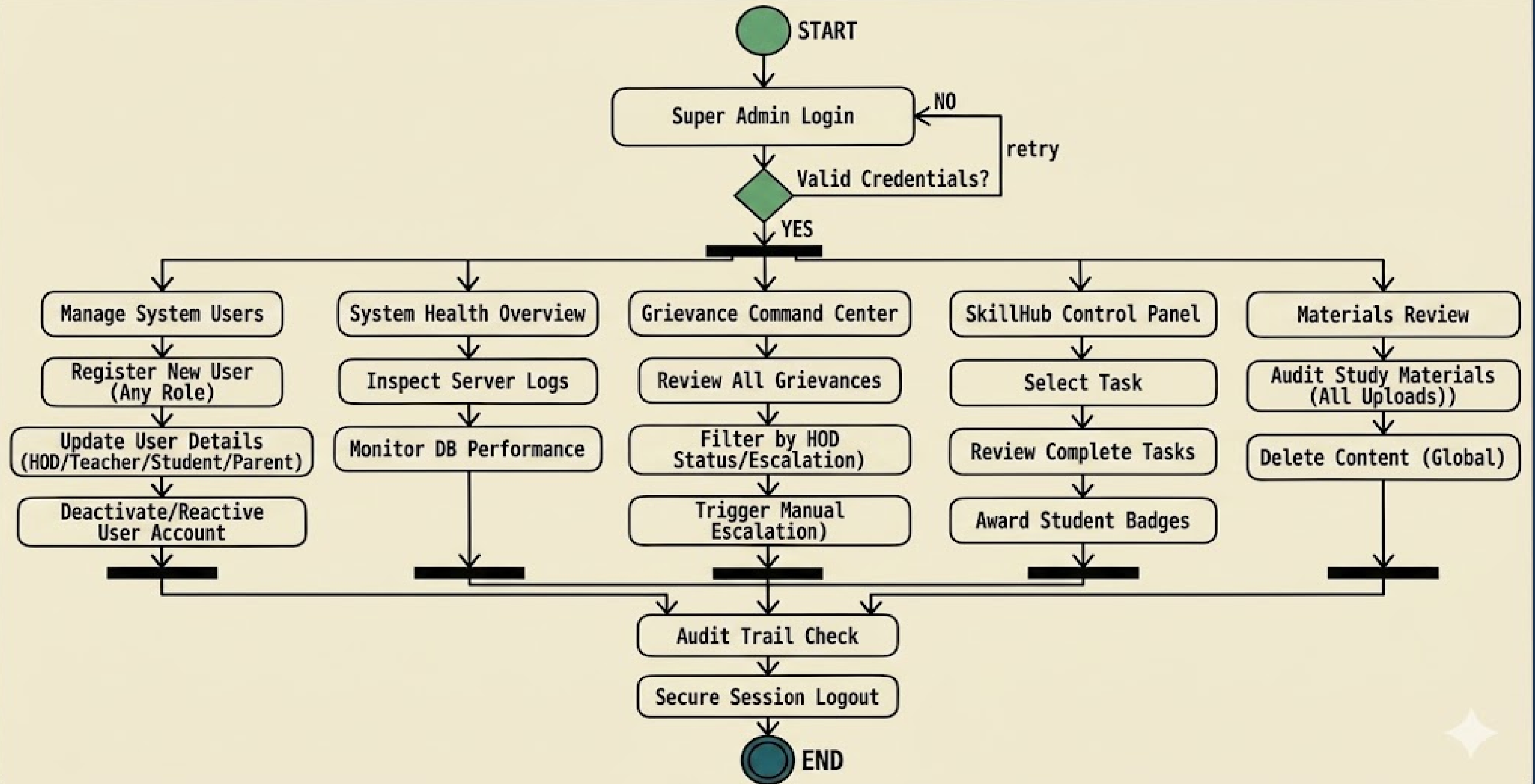
ER DIAGRAM



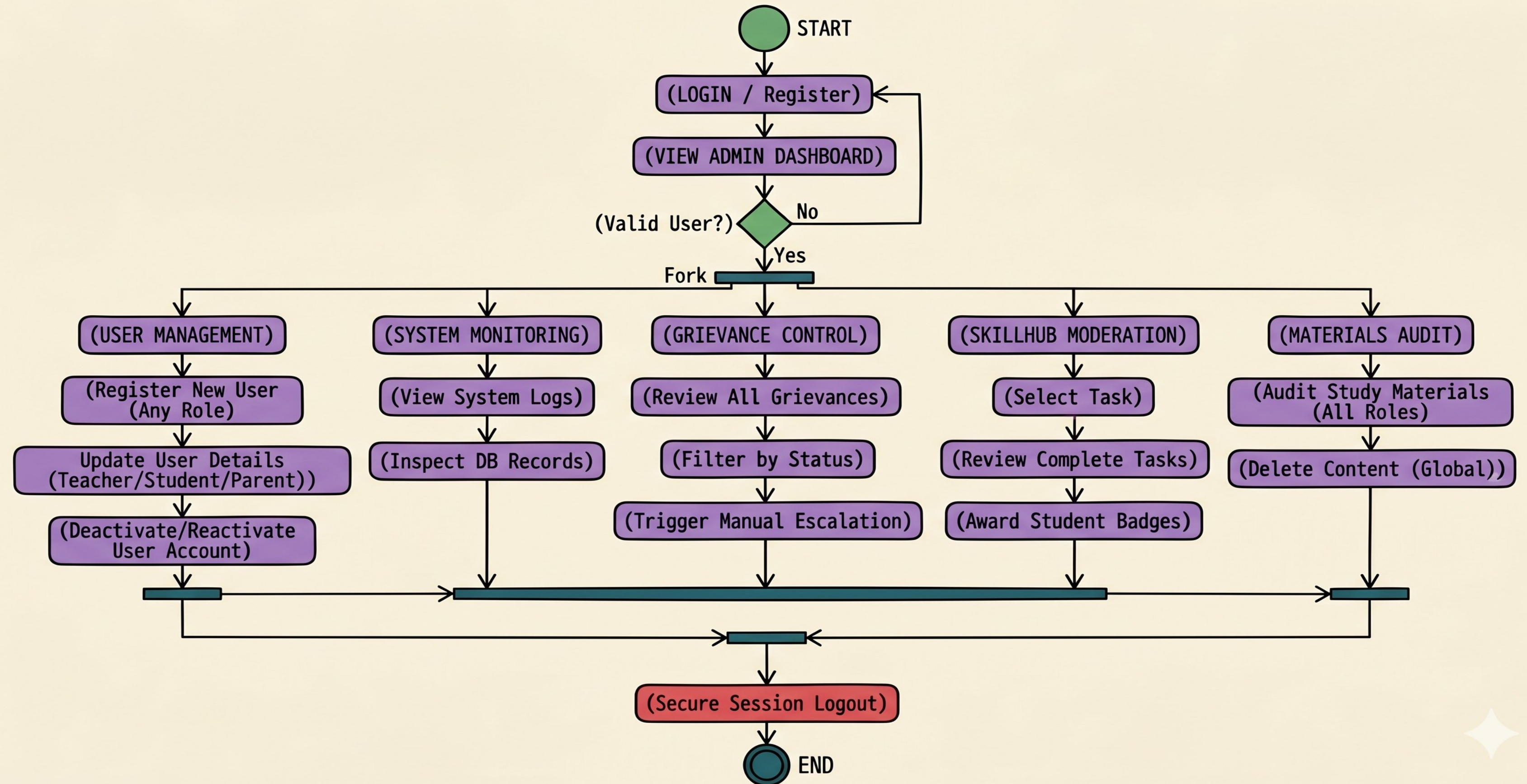
DFD – LEVEL 0 (CONTEXT DIAGRAM)



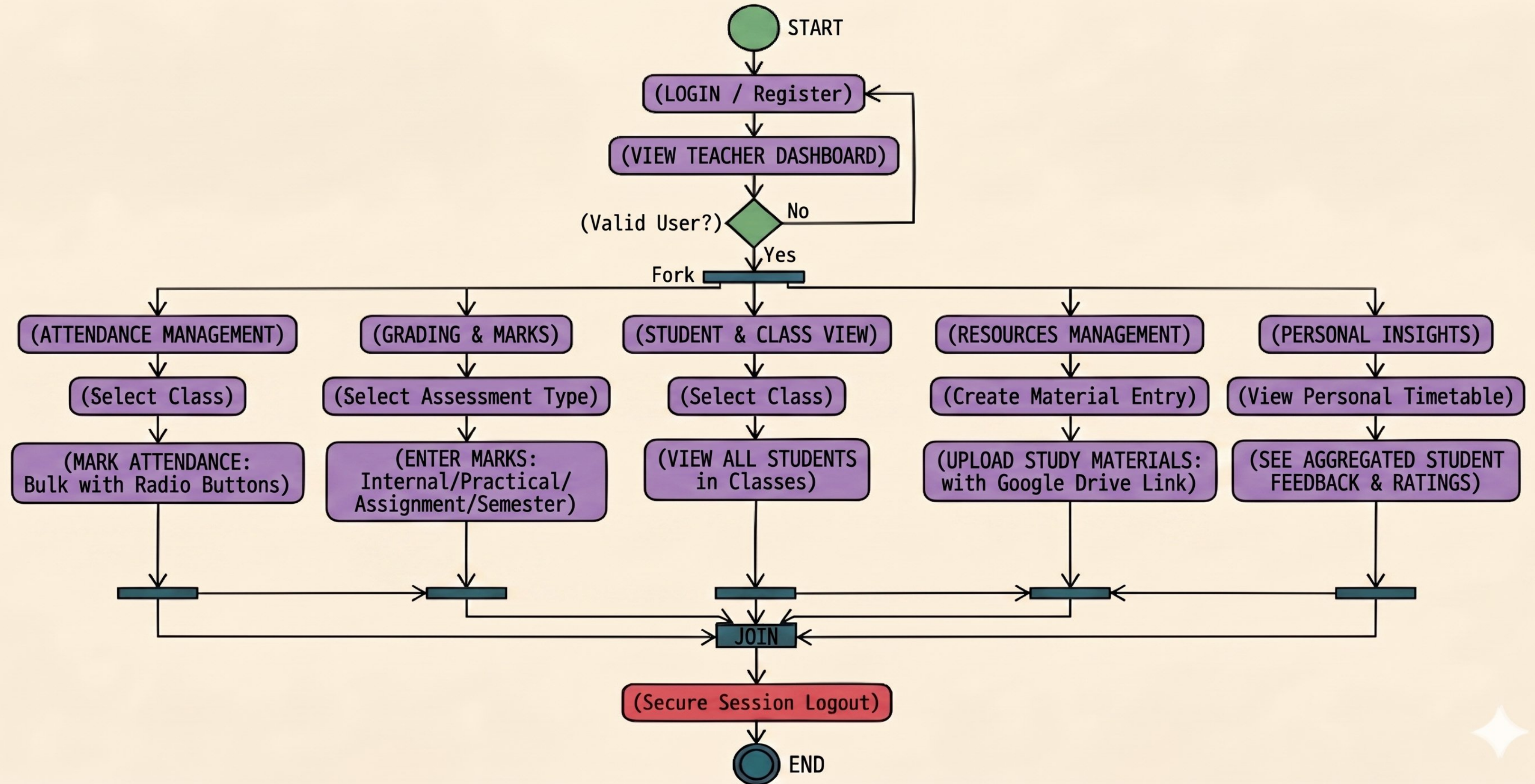
Activity Diagram for Super Admin



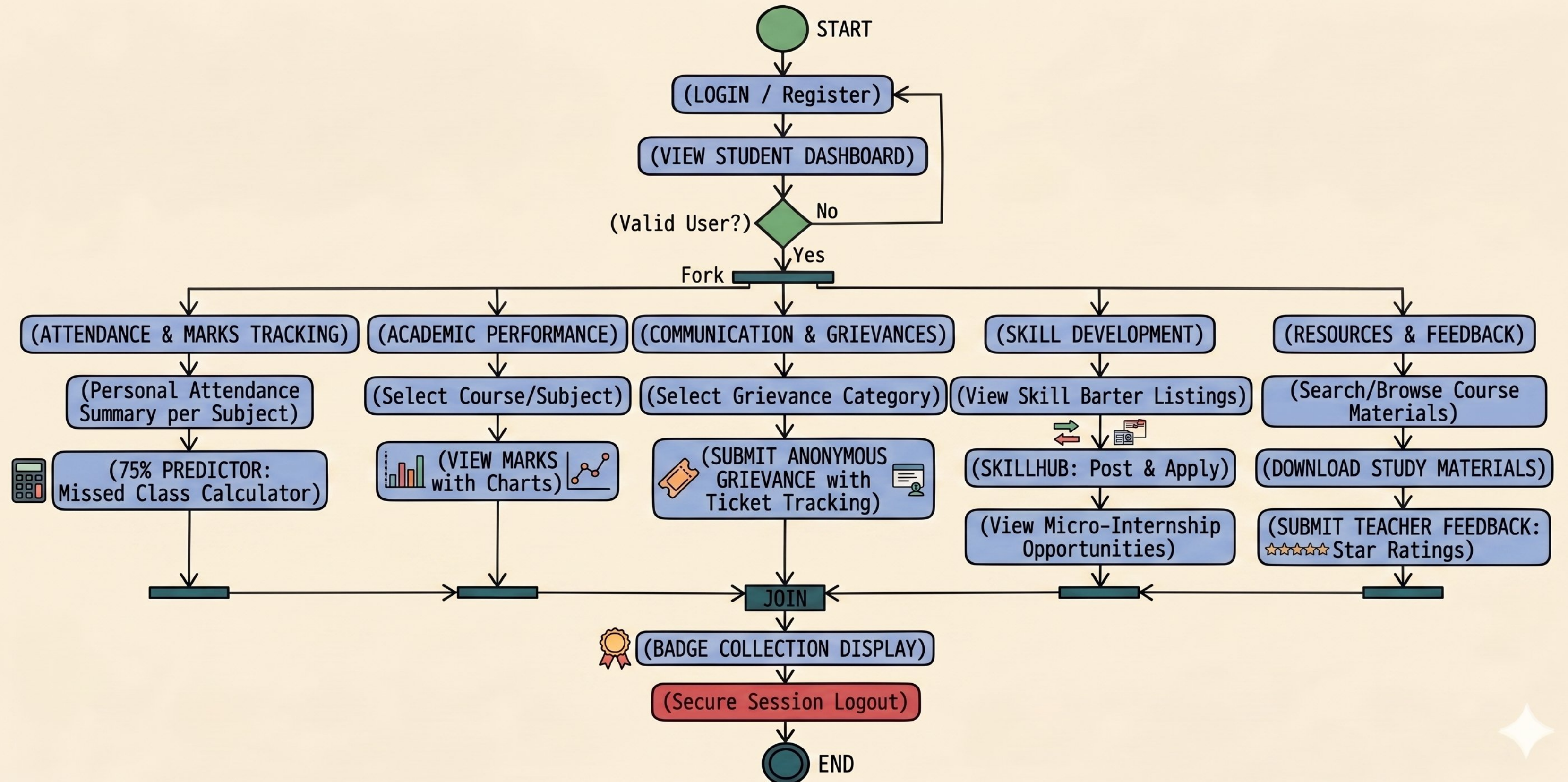
Activity Diagram for Admin



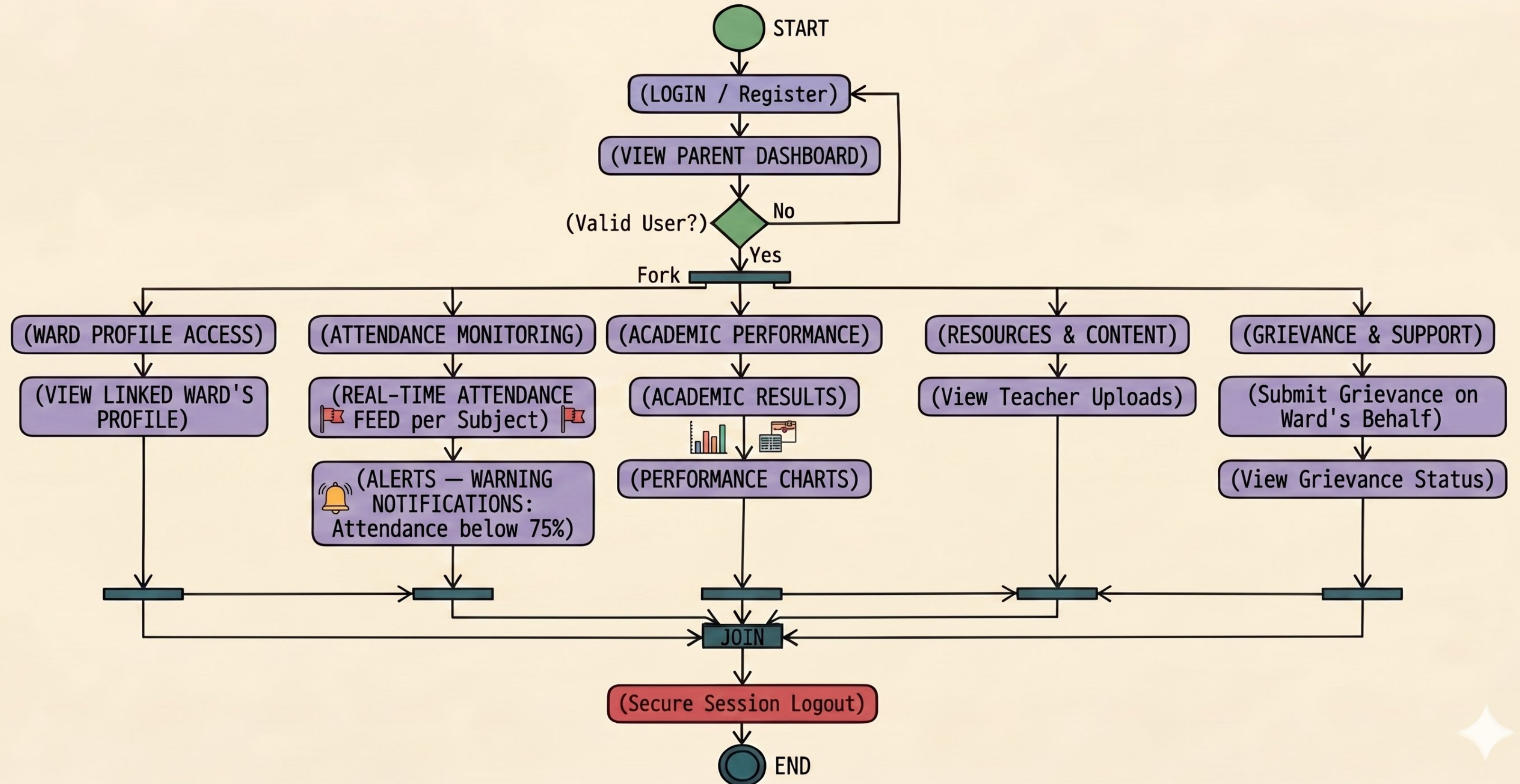
Activity Diagram for Teacher



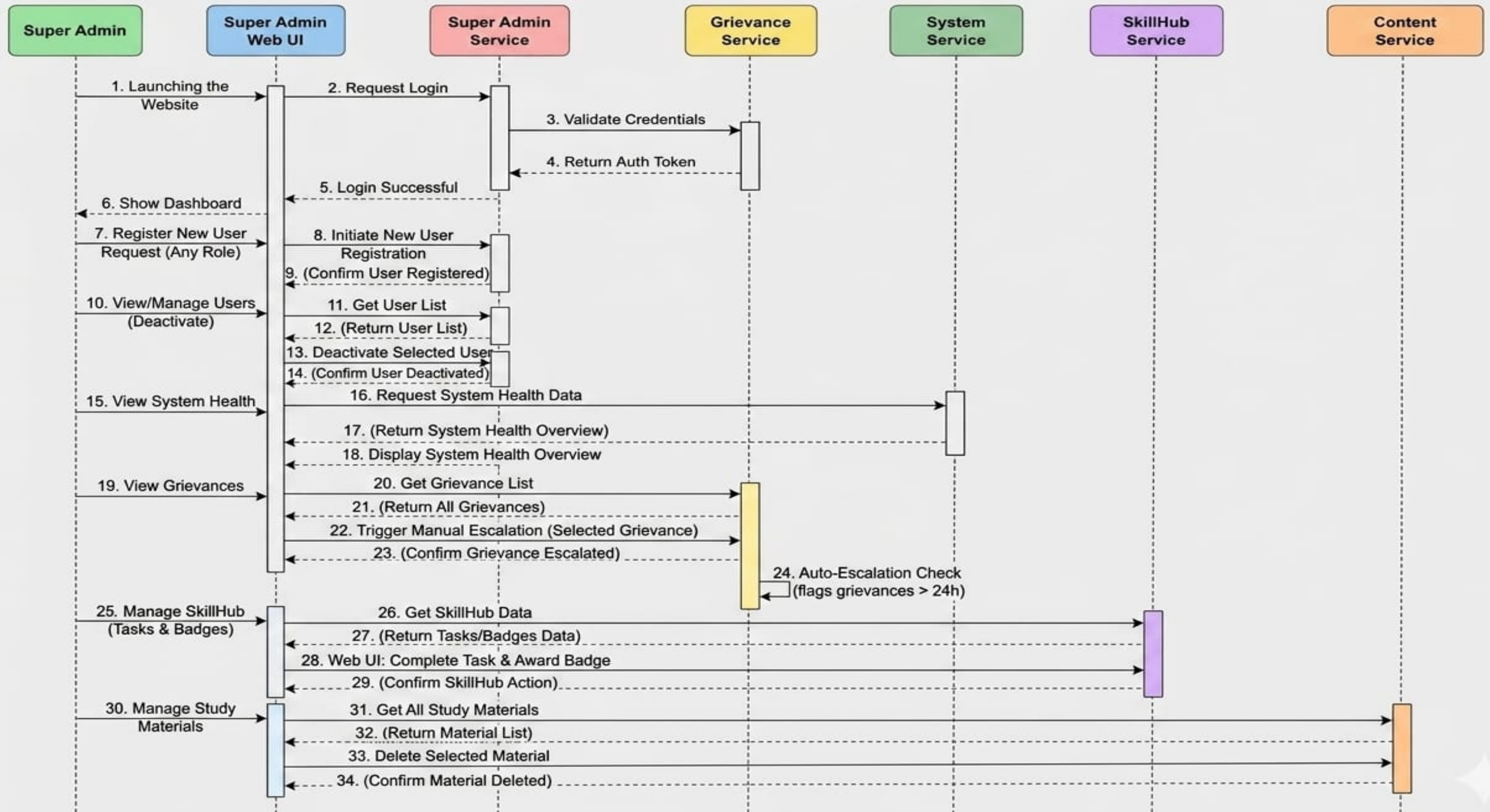
Activity Diagram for Student



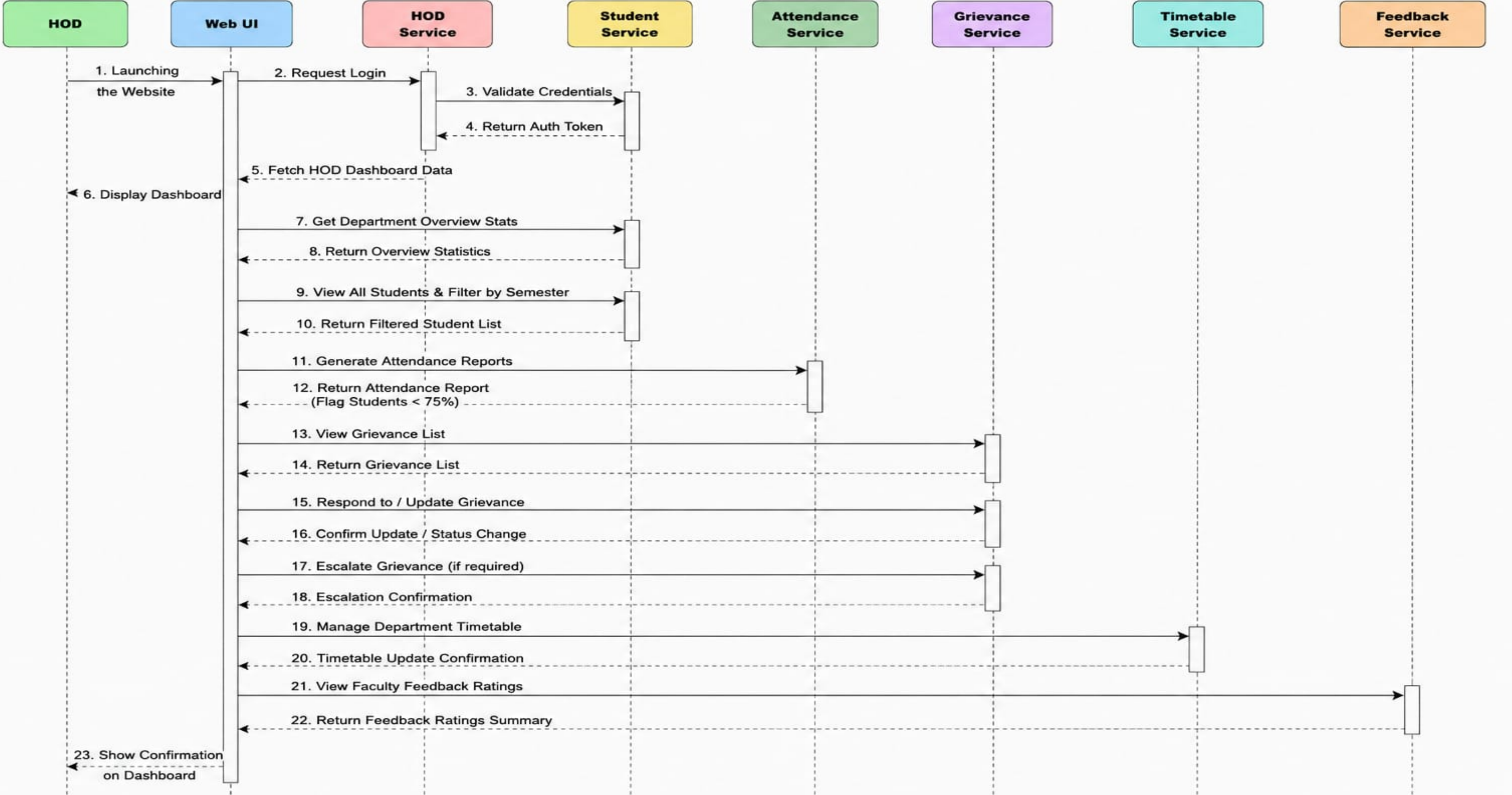
Activity Diagram for Parent



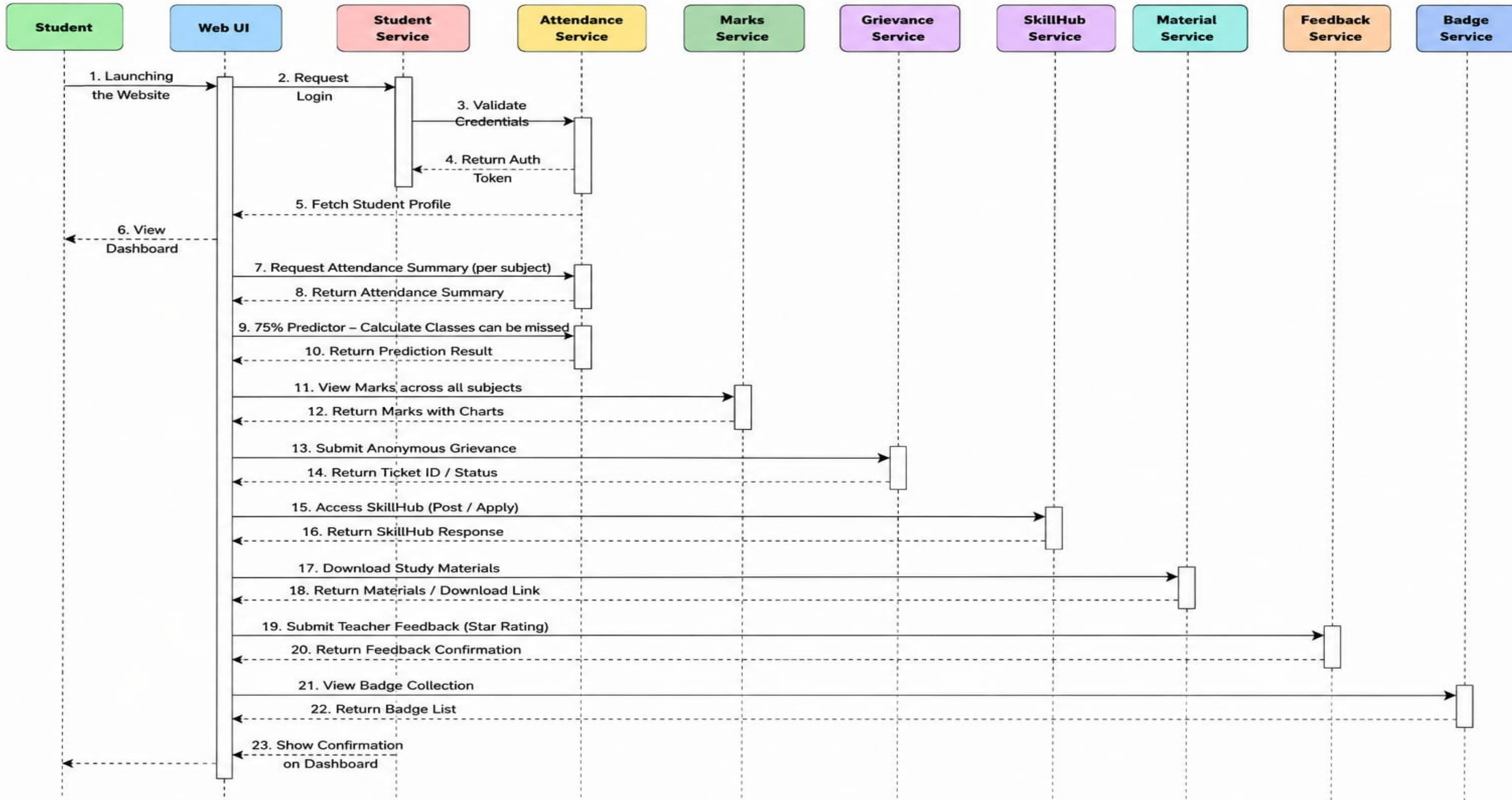
3.7.10 SEQUENCE DIAGRAM [SUPER ADMIN]



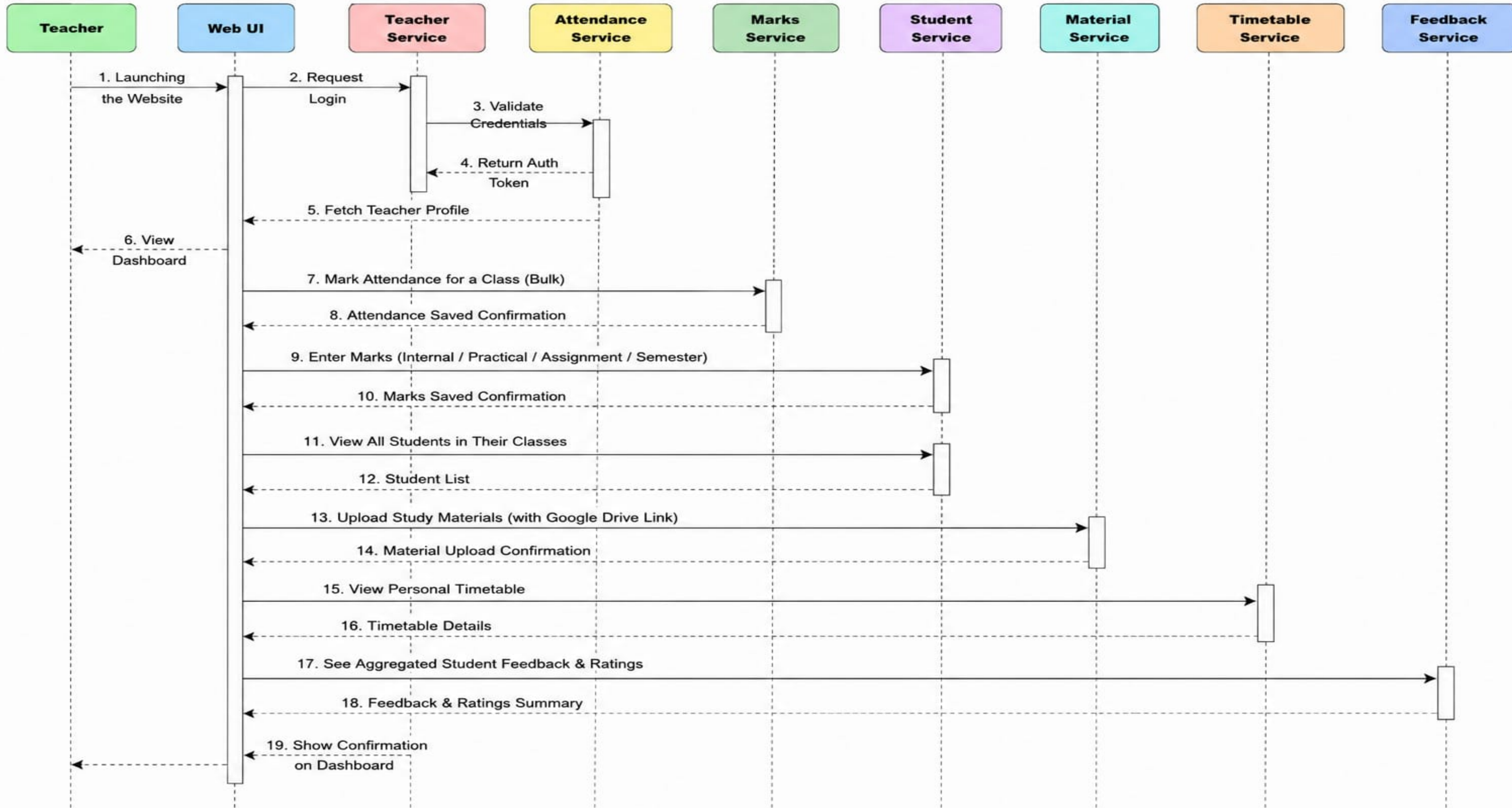
3.7.7 SEQUENCE DIAGRAM [HOD]



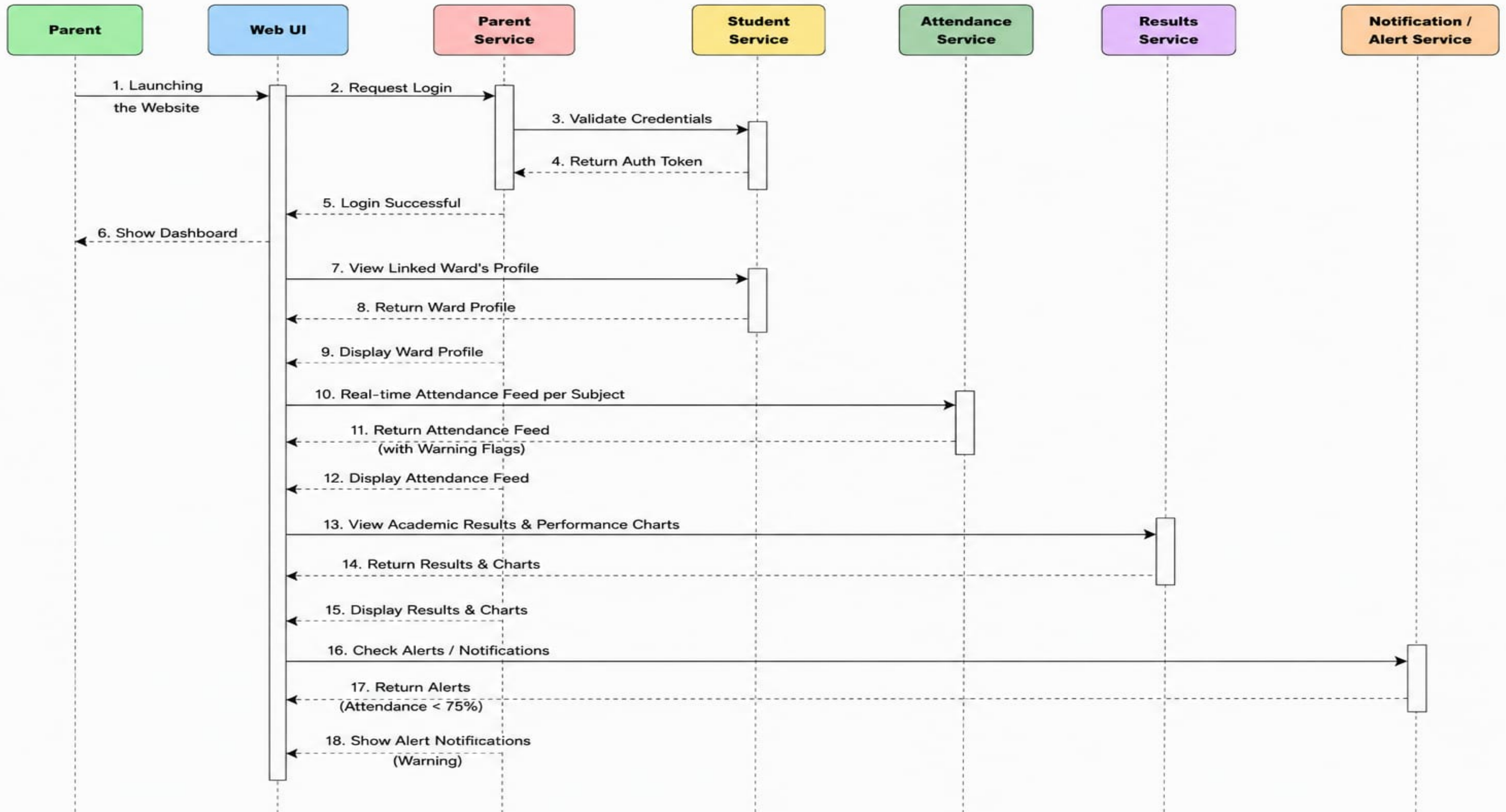
3.7.6 SEQUENCE DIAGRAM [STUDENT]



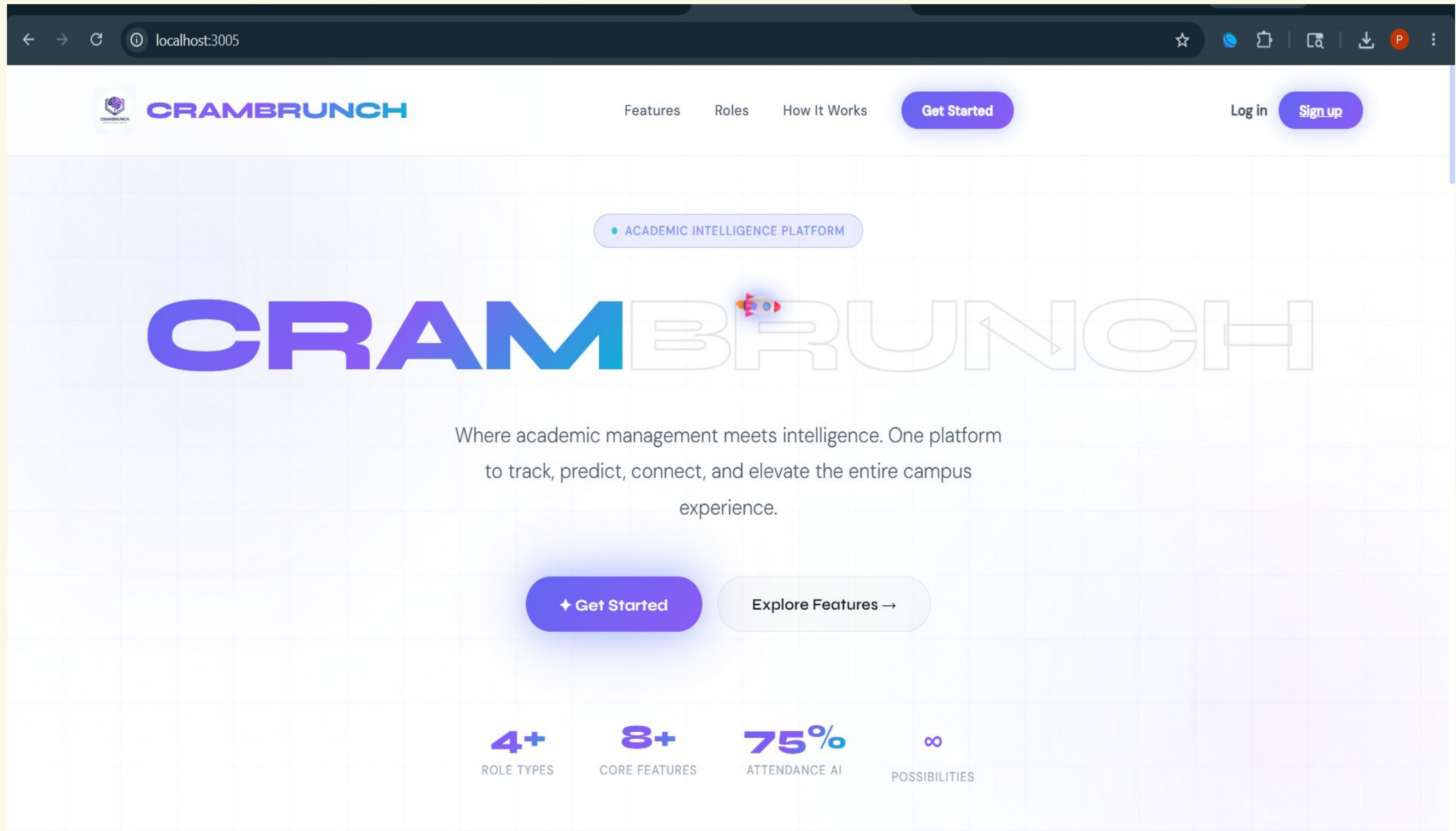
3.7.5 SEQUENCE DIAGRAM [TEACHER]



3.7.8 SEQUENCE DIAGRAM [PARENT]



Snapshots Of Project



←→↻localhost:3005



FeaturesRolesHow It Works

Get Started

Log inSign up

CORE FEATURES

Everything your campus needs, unified.

Built to eliminate paperwork and fragmentation. Every tool your institution needs in one seamless system.



Student Management

Centralized profiles for every student — demographics, academic history, and contact details, all in one secure place.



Attendance Tracking & Prediction

Auto-calculated attendance with smart 75% prediction alerts — know your exam eligibility before results day.



Performance Monitoring

Track grades, subject-wise trends, and academic growth with beautiful visual dashboards powered by Chart.js.

←→↻localhost:3005



FeaturesRolesHow It Works

Get Started

Log inSign up



Library System

Manage book inventory, issue/return cycles, and due-date tracking without a single sheet of paper.



Skill Exchange

Students can share expertise and learn from peers — a collaborative knowledge network built right into campus life.



Anonymous Complaints

Submit issues anonymously with support-based escalation, ensuring every voice is heard without fear.



Feedback & CR Selection

Transparent online CR elections and teacher feedback systems for unbiased, democratic campus governance.



Notifications & Study Materials

Auto-email parents on low attendance and result announcements. Organized study resource sharing for all.

←→↻localhost:3005



FeaturesRolesHow It Works

Get Started

Log inSign up

ROLES

Built for everyone on campus.

Role-based access ensures the right people see the right information — always.



Student

View attendance, track grades, access study materials, exchange skills, submit complaints, and vote in CR elections — all from one dashboard.



Teacher

Mark attendance, upload grades, share resources, view student performance analytics, and receive transparent peer feedback.



Parent

Receive real-time email alerts for attendance drops and result declarations. Stay connected to your child's academic journey without the guesswork.

←→↻localhost:3005



FeaturesRolesHow It Works

Get Started

Log inSign up



Administrator

Full system oversight — manage users, oversee complaints, control data access, generate reports, and configure campus-wide settings.

HOW IT WORKS

Four steps to



— HOW IT WORKS

Four steps to campus intelligence.



Login

Secure role-based authentication for students, teachers, parents, and admins.



Dashboard

A personalized command center showing your most important data at a glance.



Manage

Track attendance, upload grades, manage library records, and handle complaints effortlessly.



Insights

Visual reports and predictive analytics that drive smarter academic decisions.



— GET STARTED TODAY

Digitize Academic Management.

Join the future of campus intelligence. No paperwork. No silos. Just clarity.

✦  Start Now

×

+

Sign In

Sign Up

Student Registration

Register, verify your email, and wait for HOD approval

FULL NAME *

Your full name

EMAIL ADDRESS *

your@email.com

PASSWORD *

.....

ROLL NUMBER *

Your roll number

DEPARTMENT *

Select Department...

▼

SEMESTER *

×

+

PHONE NUMBER *

Your phone number

PARENT NAME *

Parent's full name

PARENT EMAIL *

Parent's email address

ADDRESS *

Your complete address

DATE OF BIRTH *

dd-mm-yyyy

📅

Submit for Approval

After email verification, your registration will be reviewed by the HOD.

Sign In

Sign Up

Welcome Back

Sign in to access your CRAMBRUNCH dashboard

EMAIL ADDRESS

your@email.com

PASSWORD

.....

[Forgot Password?](#)

Sign In

Reset Password

Enter your email to receive reset OTP

EMAIL ADDRESS

your@email.com

Send Reset Code

←

✦

📁

🗑️

✉️

⋮

Your OTP for CRAMBRUNCH

Inbox

CRAMBR...
to me

28 Apr

😊

↩️

⋮

CRAMBRUNCH - Email Verification

Hello,
Your OTP code for email verification is:

472501

This code will expire in 10 minutes.
If you didn't request this verification, please ignore this email.

This is a test email from CRAMBRUNCH development environment.

↩️ Reply

➡️ Forward

😊



SUPER ADMIN



Super Admin

Super Admin

CONTROL



Control Center



User Management



Departments &
HODs



Master Data

SYSTEM



Audit Ledger



All Grievances



SkillHub



Materials



Faculty Feedback

Control Center



System Control Center

Full system overview and audit dashboard



16

Total Users



4

Students



2

Open Grievances

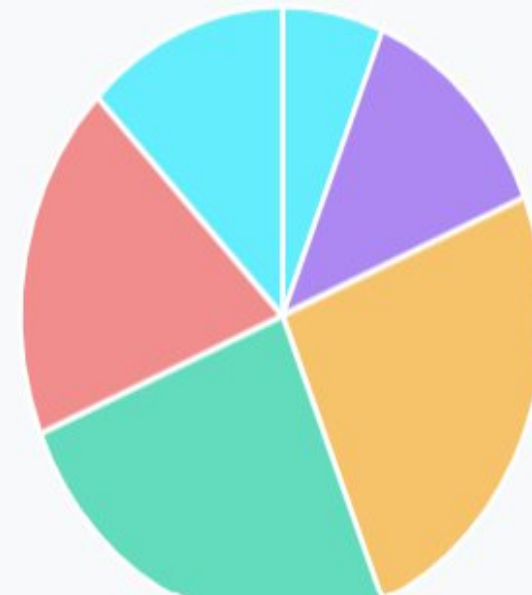


1

SkillHub Posts

User Role Distribution

librarian parent student teacher hod super_admin



HOD PORTAL

Dr. HOD

Department Head

OVERVIEW

Dashboard

DEPARTMENT

Courses & Durations

Students

Faculty

Attendance Reports

Marks Overview

ADMINISTRATION

Student Approvals

Create Teacher

Grievances

Timetable

HOD Dashboard

Computer Science

Alerts

Department Overview

Computer Science operations at a glance



2

Total Students



2

Faculty Members



1

Open Grievances



Below 75% Attendance

Grievances by Status

open

in_progress

escalated

resolved

closed



Attendance Risk Snapshot



Generate the attendance report to see the current risk list.

TEACHER PORTAL

Prof. Teacher

Teacher

OVERVIEW

Dashboard

CLASSROOM

Mark Attendance

Enter Marks

My Students

RESOURCES

Upload Materials

My Timetable

My Feedback

Micro-Internships

Logout

Dashboard

Welcome, Prof.

Your classroom overview



4

Total Students



Attendance Today



-/5

Avg Feedback Rating



1

Materials Uploaded

Today – Quick Attendance

Go to [Mark Attendance](#) to record today's class.

Go to Attendance →

Course Structure (Assigned)

STUDENT PORTAL

John Student

Student

OVERVIEW

Dashboard

My Profile

ACADEMICS

Attendance

My Marks

Timetable

Study Materials

Library

CAMPUS

Grievances

CR Selection

OPPORTUNITIES

Dashboard

Good day, John

Here's your academic snapshot



83.8%

Avg Attendance



52.7%

Avg Marks %



0

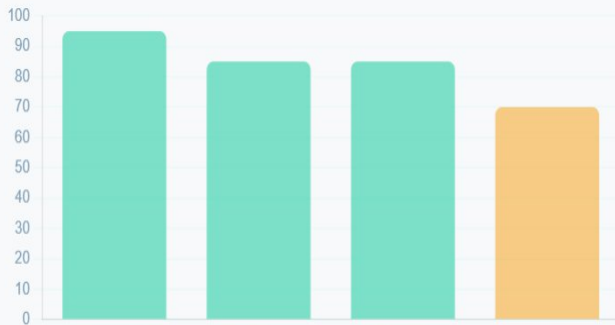
Badges Earned



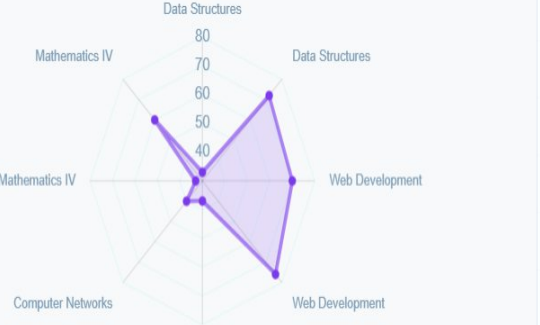
1

Open Grievances

Attendance by Subject



Performance Overview



PARENT PORTAL

Mr. Parent

Guardian

MY WARD

Overview

Attendance Feed

Academic Results

Circulars

Request Meeting

Logout

Parent Overview

John Student

Roll: CS2024001 • Computer Science • Semester 4



83.8%

Overall Attendance



52.7%

Avg Marks Score



0

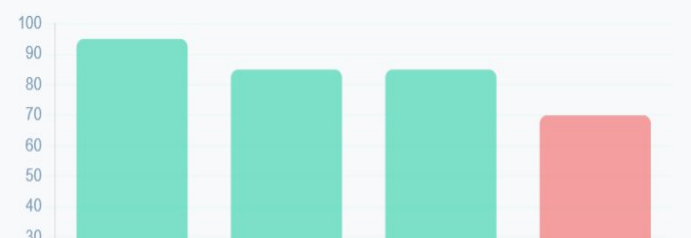
Badges Earned



At Risk

Exam Eligibility

Attendance by Subject



Academic Performance



Summary

CRAMBRUNCH is designed to revolutionize smart campus management by providing a centralized platform for academic monitoring, communication, and student engagement. The system offers role-based dashboards for Students, Teachers, Parents, HODs, Admins, and Super Admins to ensure secure and efficient management of campus activities.

Its major features include real-time attendance tracking, a 75% attendance eligibility predictor, marks management, grievance handling, study material sharing, and SkillHub activities for collaboration and skill exchange. The platform also supports parent alerts, notification systems, feedback management, and report generation for better academic transparency.

With a modern user-friendly interface and structured modules, CRAMBRUNCH improves communication, reduces manual workload, and enhances overall campus efficiency by creating a smart, connected, and digitally managed educational environment.

Future Scope - I

❖ **Smart Attendance Prediction**

The system can be enhanced to automatically predict the number of remaining classes a student must attend to maintain 75% attendance eligibility.

❖ **Mobile Application Integration**

A dedicated Android/iOS application can be developed for easier access to attendance, marks, notifications, and campus updates.

❖ **Biometric & Face Recognition Attendance**

Attendance marking can be automated using biometric systems or face recognition technology for higher accuracy and security.

Future Scope - II

❖ **Real-Time Chat & Communication**

Future upgrades may include live chat features between students, teachers, parents, and administrators for faster communication

❖ **Online Assignment & Exam Module**

Future implementation may support assignment submission, online tests, result generation, and plagiarism checking.

❖ **AI-Powered SkillHub Recommendations**

SkillHub can be improved with AI to recommend suitable projects, internships, and skill exchange opportunities based on student interests and performance.

References

Books and online study materials : For understanding software engineering concepts, database management systems, and web technologies

[W3Schools](#) : For frontend development tutorials using HTML, CSS, and JavaScript

[GeeksforGeeks](#) : For backend development concepts, Node.js, Express.js, and MongoDB references

[ChatGPT](#) : For project guidance, documentation support, coding assistance, and content generation

[Google Gemini](#) : For idea generation, research support, and technical references

[MongoDB](#) : For NoSQL database design, schema creation, and database integration

[GitHub](#) : For source code management, version control, and project collaboration

[Canva](#) : For presentation design, UI mockups, and project posters

[Draw.io](#) : For creating ER diagrams, DFDs, and UML diagrams

Thank you.



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